



Fig. 4: Loading the DS18B20 module



Fig. 5: ID of temperature sensor module

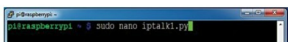


Fig. 6: Creating file iptalk1.py

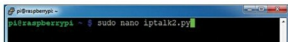


Fig. 7: Creating file iptalk2.py



Fig. 8: Running the Python code

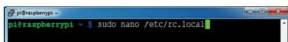


Fig. 9: Editing the rc.local file

Raspberry Pi remotely using SSH and execute all the commands directly.

After login to Raspberry Pi, load the DS18B20 module and get the temperature data ready for viewing, using the commands:

```
$ sudo modprobe w1-gpio
$ sudo modprobe w1-therm
```

You will need ID number of the DS18B20 module to read the temperature through the application program. You can get the ID using the command mentioned below, as shown in Fig. 5:

```
$ ls /sys/bus/w1/devices/
28-000004ee-228a
```

In your case, it will be similar but certainly different. Note this ID as it will be used in the application program to get the temperature data.

The application program is written using Python programming language. Download the source code (iptalk1.py and iptalk2.py) from EFY DVD or source.efymag.com.

Create a new file with the name iptalk1.py and open it in nano editor, using the command mentioned below, as shown in Fig. 6:

```
$ sudo nano iptalk1.py
```

Copy the source code from the downloaded file iptalk1.py to this new file created in Raspberry Pi. Save it by using ctrl+o and then exit through ctrl+x.

Similarly, create another file with the name iptalk2.py in the same directory and open it in nano editor, using the command mentioned below, as shown in Fig. 7:

```
$ sudo nano iptalk2.py
```

Copy the source code from the downloaded iptalk2.py file to this new file created in Raspberry Pi. In this code, you will have to make the changes for the temperature module ID noted in Fig. 5. Change the ID in the section mentioned below and save the file using ctrl+o and then exit through ctrl+x.

```
try:
    while True:
        tfile = open("/sys/bus/w1/
        devices/28-0000049582dd/w1_slave")
        text = tfile.read()
```

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on source.efymag.com.

```
tfile.close()

secondline = text.split("\n")[1]

temperaturedata = secondline.
split(" ")[9]

temperature =
float(temperaturedata[2:])

temperature = temperature / 1000

temp = 'Temp: '+str(temperature)+' C'
```

Please note, the indentation of the copied code in above two files should exactly be the same. Python is sensitive to indentation.

Finally, run the source code using the command below, as shown in Fig. 8, and you will see the LCD display shown in Fig. 1:

```
$ sudo python iptalk2.py
```

For making it work on every boot, open the '/etc/rc.local' file in nano editor, as shown in Fig. 9. Then, add the line 'sudo python /where-your-file-is/iptalk2.py' in the beginning.

Now, every time you boot your Raspberry Pi computer, the program will execute automatically and the LCD will display all the information, as shown in Fig. 1. Touch the temperature sensor and see the temperature on LCD increasing. ●

The author is an avid user of open source software. Professionally, he is a thermal power expert and works as an additional general manager at NTPC Limited



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